

Displays

D1:

$$R_{[t_0, t_1]} = \{\Phi(t_1; t_0, x_0, u, d) : x_0 \in \mathcal{X}_0, u \in \mathcal{U}, d \in \mathcal{D}\}, \quad (1)$$

D2:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\hat{R}_{[t_0, t_1]}(\theta)) \\ & \text{subject to} && P_Z(\hat{R}_{[t_0, t_1]}(\theta)) \geq 1 - \epsilon \end{aligned} \quad (2)$$

D3:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && J(\theta) \\ & \text{subject to} && P_Z(g(\theta, Z) \leq 0) \geq 1 - \epsilon \\ & && \theta \in \Theta, \end{aligned} \quad (3)$$

D4:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && J(\theta) \\ & \text{subject to} && g(\theta, z^{(i)}) \leq 0, \quad i = 1, \dots, N \\ & && \theta \in \Theta, \end{aligned} \quad (4)$$

D5:

$$N \geq \frac{1}{\epsilon} \left(\frac{e}{e-1} \right) \left(\log \frac{1}{\delta} + n_\theta \right), \quad (5)$$

D6:

$$\hat{R}_{[t_0, t_1]}(A, b) = \{x : \|Ax - b\|_p \leq 1\} \quad (6)$$

D7:

$$\begin{aligned} & \arg \min_{A, b} && -\log \det A \\ & \text{s.t.} && P_Z(\|AZ - b\|_p - 1 \leq 0) \leq 1 - \epsilon, \end{aligned} \quad (7)$$

D8:

$$P_S(P_Z(\hat{R}_{[t_0, t_1]}(A, b)) \geq 1 - \epsilon) \geq 1 - \delta; \quad (9)$$

D9:

$$P_S(\hat{R}_{[t_0, t_1]}(A, b) \text{ is } \epsilon\text{-accurate with respect to } Z) \geq 1 - \delta. \quad (10)$$

D10:

$$\begin{aligned} & \arg \min_{A, b} && -\log \det A \\ & \text{subject to} && \|Az^{(i)} - b\|_p - 1 \leq 0, \quad i = 1, \dots, N \end{aligned} \quad (8)$$

D11:

$$N_{diag} = \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(\log \frac{1}{\delta} + 2n \right) \right\rceil, \quad (11)$$

Inline

I1: $O(n^2)$

I2: $O(n)$

I3: n

I4: $\Phi(t_1; t_0, x_0, u, d)$

I5: $x_0 \in \mathbb{R}^n$

I6: t_0

I7: t_1

I8: $u : [t_0, t_1] \rightarrow \mathbb{R}^p$

I9: $d : [t_0, t_1] \rightarrow \mathbb{R}^w$

I10: $\dot{x}(t) = f(t, x(t), u(t), d(t))$

I11: $[t_0, t_1]$

I12: $\phi(t_1)$

I13: ϕ

I14: $\phi(t_0) = x_0$
 I15: $\mathcal{X}_0 \subset \mathbb{R}^n$
 I16: $\mathcal{U}$
 I17: $\mathcal{D}$
 I18: $\mathcal{X}_0$
 I19: $\hat{R}_{[t_0, t_1]}$
 I20: θ
 I21: $\hat{R}_{[t_0, t_1]}(A, b) = \{x : \|Ax - b\|_2 \leq 1\}$
 I22: $\theta = (A, b)$
 I23: $R_{[t_0, t_1]} \subset \hat{R}_{[t_0, t_1]}$
 I24: $\hat{R}_{[t_0, t_1]} \subset R_{[t_0, t_1]}$
 I25: Z
 I26: p_Z
 I27: $p_Z(x) = 0$
 I28: x
 I29: $p_Z(x) > 0$
 I30: $R_{[t_0, t_1]}$
 I31: $P_Z(\hat{R}_{[t_0, t_1]}) \geq 1 - \epsilon$
 I32: P_Z
 I33: ϵ
 I34: $1 - \epsilon$
 I35: J
 I36: g
 I37: $\Theta \in \mathbb{R}^{n_\theta}$
 I38: $\{z^{(i)}\}_{i=1}^N$
 I39: N
 I40: $\delta \in (0, 1)$
 I41: e
 I42: $\geq 1 - \delta$
 I43: p
 I44: $x, b \in \mathbb{R}^n$
 I45: $A \in \mathbb{R}^{n \times n}$
 I46: ≥ 1
 I47: ∞
 I48: $p = 2$
 I49: $p = \infty$
 I50: A
 I51: $-\log \det A$
 I52: $\hat{R}_{[t_0, t_1]}(A, b)$
 I53: $-\log \det(A)$
 I54: $\|AZ - b\|_p - 1$
 I55: b
 I56: Φ
 I57: X_0
 I58: U
 I59: D
 I60: $Z = \Phi(t_1; t_0, X_0, U, D)$
 I61: $\epsilon, \delta \in (0, 1)$
 I62: P_S
 I63: $N = \left\lceil \frac{1}{\epsilon} \frac{e}{e-1} \left(\log \frac{1}{\delta} + n(n+1)/2 + n \right) \right\rceil$
 I64: (A, b)
 I65: $n \times n$
 I66: $n_\theta = n(n+1)/2 + n$
 I67: n_θ
 I68: $1 - \delta$
 I69: $\blacksquare$

I70: δ
 I71: $\hat{R}_{[t_0, t_1]}(A, b) = \{x : \|Ax + b\|_p \leq 1\}$
 I72: $i \in \{1, \dots, N\}$
 I73: $x^{(i)}$
 I74: $u^{(i)}$
 I75: $d^{(i)}$
 I76: $z^{(i)} = \Phi(t_1; t_0, x^{(i)}, u^{(i)}, d^{(i)})$
 I77: $O(\frac{1}{\epsilon})$
 I78: $O(\log \frac{1}{\delta})$
 I79: $n(n+1)/2$
 I80: $n_\theta = 2n$
 I81: $\mathbb{R}^n$
 I82: $z^{(i)}$
 I83: $\epsilon = 0.05$
 I84: $\delta = 10^{-9}$
 I85: 0.95
 I86: $N = 1510$
 I87: $N_{diag} = 1036$
 I88: y
 I89: 46,052
 I90: .01
 I91: 0.9999
 I92: $t = 0$
 I93: $t = 1.75$
 I94: $t = 3.5$